12. option(a)

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[A_0]}{[A]}\right)$$

$$2.303 \times 10^{-2}\ \text{hour}^{-1} = \left(\frac{2.303}{1806\ \text{min}}\right)\log\left(\frac{0.25}{[A]}\right)$$

$$\left(\frac{2.303 \times 10^{-2}\ \text{hour}^{-1} \times 1806\ \text{min}}{2.303}\right) = \log\left(\frac{0.25}{[A]}\right)$$

$$\left(\frac{1806 \times 10^{-2}}{60}\right) = \log\left(\frac{0.25}{[A]}\right)$$

$$0.301 = \log\left(\frac{0.25}{[A]}\right)$$

$$\log 2 = \log\left(\frac{0.25}{[A]}\right)$$

$$2 = \left(\frac{0.25}{[A]}\right)$$

$$[A] = \left(\frac{0.25}{2}\right) = 0.125\text{M}$$

13. option(b)

$$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[A_0]}{[A]}\right)$$

$[A_0] = 100$; $[A] = 25$

$$6.909 = \left(\frac{2.303}{t}\right)\log\left(\frac{100}{25}\right)$$

$$t = \left(\frac{2.303}{6.909}\right)\log(4)$$

$$t = \left(\frac{1}{3}\right)\log 2^2$$

$$t = \left(\frac{2}{3}\right)\log 2$$

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